

MINISTRY OF INDUSTRY AND MINERAL RESOURCES

MVP1 IPAD / INSPECTOR WORKSPACE

FUNCTIONAL SPECIFICATION

Field execution, pre-start, offline package, journey telemetry, geofence, inspection, evidence, sync, submission and correction.

Document code	Version	Date
MIM-INSP-MVP1-IPAD-FS	2.0 - MVP1 Final Documentation Contract	10 July 2026

BINDING DELIVERY RULE

MVP1 is governed by the complete 9-tab, 478-row customer/auditor spreadsheet baseline. No requirement may be silently removed, weakened, generalized away, replaced by mock behavior, or deferred to MVP2. Strategic BRD content is used only to close implementation-critical gaps and strengthen integrity, auditability, security, offline resilience and operability.

Primary audience	Status
Business, Design, Engineering, QA and Fable/Claude implementation teams	Authoritative implementation input - use with the MVP1 acceptance workbook

1. Purpose, authority and scope boundary

This specification is the independently buildable functional contract for the MVP1 iPad / Inspector Workspace. It must be used together with the enhanced MVP1 BRD and the MVP1 Breakdown and Acceptance Contract workbook. The primary functional authority remains all applicable rows from the 478-row Inspection Project.xlsx baseline.

Code	Module	Rows	Mandatory scope
M03	Inspection Execution - Pre-Start	15	Assigned visits, schedule views, preparation, Factory 360 snapshot, execution eligibility, package readiness, appointments and assignment return.
M04	Physical Inspection Execution	223	Package download, journey telemetry, travel, arrival, geofence, execution workspace, checklist, report, evidence, violations, action forms, submission and audit.

No requirement mapped to this channel may be moved to MVP2, reduced to a static mock, or declared complete without runtime and evidence acceptance.

2. Personas, responsibilities and information architecture

Persona	Primary responsibilities
Inspector	Own field preparation, journey, check-in, physical execution, evidence, exception handling, submission and correction.
Factory Representative	Provide access, verify identity/authority, support physical inspection and acknowledge/sign only where approved flow requires.
Operations User	Monitor active journey/execution and exceptions without overriding field evidence silently.
System	Resolve package, enforce configuration, persist offline work, sync safely, validate submission and maintain audit/version lineage.

Mandatory navigation and screen surfaces

Screen / surface	Purpose and mandatory behavior
Assigned visits dashboard	Today/upcoming/completed/readiness status with fast access to next visit.
Calendar / list / map	Field-friendly schedule and route views.
Visit card and Factory 360 snapshot	Key factory, appointment, risk/findings and package context.
Prepare inspection	Pre-start checklist, package/documents, device/network and eligibility.
Journey session	Start journey, route, ETA, distance, deviation and live telemetry.
Arrival and geofence check-in	Official/observed location, accuracy, arrival radius, check-in and override/exception.
Inspection workspace	Dynamic report, checklist, factory verification, progress and autosave.
Evidence capture	Photo/video/document/audio with item linkage, metadata and chain of custody.
Violations / penalties / action forms	Configured runtime outcomes, corrective forms, owner and due date.
Pre-submit review	Mandatory field/evidence/form/signature blockers.
Final submission	Create immutable version, report/IDs, audit and review handoff.
Returned correction	Selective unlock, correction, new evidence, new version and resubmission.
Sync / recovery	Offline cache, local drafts, queued operations, reconnect, conflict and recovery state.

3. Field user journeys

SB06 - Inspector Pre-Start Workspace - Assignments, Preparation and Readiness

Assigned visits, views, preparation, Factory 360 snapshot, execution mode, package resolution, notifications and return assignment.

Step	Field slice	Inspector/system behavior	Output
1	Assigned Visits Dashboard	Review today/upcoming/completed assignments and readiness.	Assignment list
2	Calendar / List / Map	Choose field-friendly view of schedule and route.	Schedule view
3	Search / Filter / Sort	Find visits by factory, ID, status, date or priority.	Filtered list
4	Visit Card Summary	See visit, factory, time, location, package and status.	Visit card
5	Prepare Inspection	Complete pre-start checklist and required documents.	Preparation record
6	Factory 360 Snapshot	Review recent findings, risk and factory context.	Factory snapshot
7	Execution Mode Eligibility	Determine physical or virtual eligibility.	Eligibility decision
8	Package Resolution	Verify complete, current and offline-ready package.	Package validation
9	Notifications & Appointment	See confirmations, reminders and critical alerts.	Notification state
10	Return Assignment	Return with valid reason, notes and audit.	Return/reassignment request

SB07 - Physical Inspection Start Journey - Travel, Check-In and Session Initiation

Package integrity, journey start, device/network, navigation, live telemetry, arrival, geofence, inspection start and exceptions.

Step	Field slice	Inspector/system behavior	Output
1	Accept Assigned Visit	Review summary and accept assignment.	Accepted visit
2	Package Integrity Check	Download encrypted package and verify integrity.	Locked offline package
3	Start Journey Session	Create active journey and timestamp.	Journey session
4	GPS / Device / Network	Verify accuracy, battery, connectivity and offline readiness.	Readiness telemetry
5	Navigation & Route Guidance	Receive route, ETA and remaining distance.	Route plan
6	Live Journey Progress	Track ETA, distance and route deviation.	Live telemetry
7	Arrival Detection	Detect proximity and arrival confidence.	Arrival event
8	Geofence Check-in	Compare official/observed location and record check-in.	Check-in evidence
9	Start Inspection Session	Pass mandatory validations and initialize workspace.	Inspection session
10	Handle Exceptions	Govern cancellation, override and inability-to-execute paths.	Exception record

SB08 - Physical Inspection Execution Workspace - Report, Checklist, Evidence and Submission

Factory verification, dynamic report, checklist, violations, penalties, evidence, action forms, notes, review and submission.

Step	Field slice	Inspector/system behavior	Output
1	Inspection Workspace	Show visit, factory, scope, progress and offline/autosave state.	Inspection session
2	Factory Verification	Verify factory contact/representative and capture timestamp/location.	Verification record
3	Dynamic Report Form	Render package-defined report fields and context.	Report draft
4	Inspection Checklist	Execute sectioned items with required/optional/conditional behavior.	Checklist responses
5	Violations & Penalties	Apply configured rules and surface resulting enforcement outcomes.	Violation/penalty records
6	Evidence Capture	Capture photo/video/document/audio with item linkage and metadata.	Evidence chain
7	Action Forms	Open required corrective/legal/action forms from rules.	Action forms

Step	Field slice	Inspector/system behavior	Output
8	Notes / Observations	Record timestamped notes linked to items/forms.	Observation log
9	Pre-Submit Review	Identify incomplete mandatory data, evidence and forms.	Validation summary
10	Final Validation & Submission	Create immutable submitted version and handoff to review.	Submitted version

SB14 - Journey Telemetry, Location and Exception Control

ETA, remaining distance, route deviation, arrival radius, GPS accuracy, check-in evidence, overrides and exception handling.

Step	Field slice	Inspector/system behavior	Output
1	Journey Start	Create journey session for accepted visit.	Journey session
2	Device Readiness	Capture device, GPS, network and offline state.	Device telemetry
3	Route Guidance	Plan route and expected journey metrics.	Route plan
4	ETA & Remaining Distance	Update progress continuously.	Journey metrics
5	Route Deviation	Detect deviation from planned route.	Deviation event
6	Arrival Radius	Detect expected arrival proximity.	Arrival event
7	Geofence Entry	Validate boundary and location accuracy.	Geofence event
8	Check-in Evidence	Record official/observed coordinates, time and device.	Check-in record
9	Override / Exception	Govern low accuracy, GPS override or inability to execute.	Exception/override record
10	Operations Synchronization	Expose current state and history to Operations Center.	Operational telemetry

SB15 - Evidence, Violations, Penalties and Action Forms

Capture rules, chain of custody, auto-generated violations, penalty visibility, corrective/action forms and review traceability.

Step	Field slice	Inspector/system behavior	Output
1	Capture Response	Inspector records checklist/report result.	Response record
2	Evaluate Evidence Rule	System determines mandatory/optional evidence.	Evidence requirement
3	Capture Evidence	Create linked evidence with metadata and content identity.	Evidence record
4	Generate Finding	Create finding/observation where required.	Finding record
5	Trigger Violation	Apply published violation rules.	Violation record
6	Resolve Penalty	Apply configured penalty relationship/range.	Penalty record
7	Open Action Form	Create mandatory corrective/legal/action form.	Action form
8	Track Owner / Due Date	Assign responsibility and priority.	Action tracking
9	Review Traceability	Reviewer sees full response-evidence-violation-action lineage.	Review chain
10	Factory 360 History	Accepted outcomes update longitudinal factory dossier.	Historical record

SB16 - Inspection Submission, Locking, Version Creation and Handoff

Pre-submit validation, signatures where applicable, immutable submitted version, audit and handoff to Level 2 review.

Step	Field slice	Inspector/system behavior	Output
1	Completion Summary	Show progress and unresolved mandatory gaps.	Completion status
2	Mandatory Field Validation	Validate report/checklist values.	Validation result
3	Evidence Validation	Validate required evidence and item linkage.	Evidence result
4	Action Form Validation	Require all blocking forms complete.	Action result

Step	Field slice	Inspector/system behavior	Output
5	Signature / Acknowledgement	Capture only where required by approved MVP1 flow.	Signature record
6	Final Confirmation	Inspector confirms submission intent.	Submission confirmation
7	Create Immutable Version	Lock exact report/checklist/evidence/config versions.	Submitted version
8	Generate Report / IDs	Generate submission/report identifiers.	Report artifact
9	Audit Submission	Record actor, device, time, state, package/version references.	Submission audit
10	Handoff to Level 2	Create review queue item and notifications.	Review-ready case

SB17 - Returned Inspection Correction, Resubmission and Version Comparison

Returned section targeting, selective unlock, immutable reviewer comments, new versions and before/after comparison.

Step	Field slice	Inspector/system behavior	Output
1	Receive Return Decision	Inspector receives immutable reviewer comments and reason.	Return notice
2	Identify Affected Sections	Show only sections/items targeted for correction.	Return scope
3	Lock Unaffected Content	All other submitted content remains read-only.	Locked baseline
4	Edit Returned Sections	Inspector corrects only allowed scope.	Corrected draft
5	Add New Evidence / Forms	Add new artifacts without rewriting original history.	Correction evidence
6	Validate Corrections	Re-run mandatory rules on returned scope.	Correction validation
7	Create New Version	Resubmission creates immutable new version.	Version N+1
8	Compare Versions	Highlight changes by section/item/evidence.	Version diff
9	Reopen Review	Reviewer evaluates latest version with prior history visible.	Re-review state
10	Preserve Timeline	All versions, comments and decisions remain immutable.	Audit timeline

SB19 - MVP1 Web Portal and Inspector Workspace Information Architecture

Navigation structure connecting planning, management, Factory 360, Operations Center, review, pre-start and execution.

Step	Field slice	Inspector/system behavior	Output
1	Web Portal Home	Access planning, management, Factory 360, review and operations.	Web navigation
2	Planning Workspace	Bulk/Single/Immediate planning.	Visit plans
3	Visit Management	Operational control and bulk actions.	Managed visits
4	Factory 360	Unified factory dossier.	Factory context
5	Review Workspace	Level 2 queue, decisions and version comparison.	Review decisions
6	Operations Center	Live map, SLA, risk and exceptions.	Operational visibility
7	Inspector Home	Assigned visits and readiness.	Inspector assignments
8	Journey & Check-in	Travel, telemetry, arrival and geofence.	Journey records
9	Inspection Workspace	Report, checklist, evidence and actions.	Inspection content
10	Submission & Correction	Submit, receive returns, correct and resubmit.	Versioned submissions
11	Shared Search / Alerts / Timeline	Consistent cross-module access patterns.	Shared UX
12	Cross-Channel Deep Links	Preserve object, version and context across channels.	Context continuity

4. Offline, recovery and sync contract

- Required inspection package must be available locally before the inspector begins any journey or inspection state that depends on offline execution.
- Every field edit must autosave locally with explicit sync state. A crash/restart must recover the last durable local draft without data loss.
- Reconnect must replay queued operations idempotently; retries must not duplicate evidence, findings, violations, notifications or submissions.
- When server and local versions conflict, the system must create an explicit conflict state and resolution path. Silent overwrite is prohibited.
- Evidence captured offline must retain local identity and metadata until durable server confirmation; the UI must never claim full submission while mandatory artifacts remain unsynced.
- The inspector must always be able to distinguish offline, pending sync, syncing, synced, conflict and failed states.

5. GIS, geofence and telemetry contract

- Capture official and observed location separately; never overwrite official master coordinates with field observation without governed process.
- Record GPS accuracy, device, timestamp and configured geofence/arrival threshold or version used for the decision.
- Journey state, ETA, remaining distance and route deviation must be distinct from business workflow status.
- Check-in must require geofence/threshold validation or a governed override with reason, authority and audit.
- Operations Center receives live or delayed telemetry with freshness indication; stale data must not be presented as live.
- Location retention and visibility must follow approved policy and least privilege.

5. Engine and service dependencies

ID	Engine	Consumers	Mandatory behavior	Acceptance proof	Forbidden shortcut
ENG-01	Regulatory & Compliance Engine	Web, iPad, Virtual, Review	Manage regulations, item catalogue, applicability, response semantics, dependencies and published runtime versions.	Publish configuration, execute an inspection against it, then prove the submission references the same version	Hard-coded checklist detached from admin configuration
ENG-02	Dynamic Form & Package Builder	iPad, Virtual, Review	Build report packages, sections, items, required/optional/conditional behavior, action forms and evidence requirements.	Change a draft rule, publish a new version, prove old inspection remains on old version and new inspection uses new version	Static forms coded screen-by-screen
ENG-03	Workflow Engine	All channels	States, transitions, allowed actors, guards, required fields, side effects, notifications, SLA and escalation.	Demonstrate configurable transition and prove runtime follows it	Hard-coded status chips called a workflow engine
ENG-04	Risk Engine	Planning, Factory 360, Operations, Review where applicable	Risk variables, weights/thresholds, source provenance, score calculation, version reference and explainability sufficient for MVP1 targeting/visibility.	Recalculate a factory score from stored inputs and version; show planning/Factory 360 consumption	Decorative risk gauge with no reproducible logic
ENG-06	GIS, Geofence & Journey Telemetry	Planning, iPad, Operations Center	Official/observed location, geofence, GPS accuracy, journey state, ETA/distance, route deviation, arrival radius, check-in, override and audit.	Complete journey from start to geofence and show synchronized Ops Center evidence	Map widget without governed location events
ENG-07	Evidence & Media Chain of Custody	iPad, Virtual, Review, Factory 360	Evidence capture rules, metadata, hashing/identity, item linkage, time/location/device attribution, upload/sync, immutable submitted reference.	Capture offline and online evidence, sync, review, and prove chain of custody	Loose file upload without item linkage and immutable history
ENG-08	Violation & Penalty Engine	iPad, Review, Factory 360, Operations	Violation catalogue, severity/level, response-to-violation rules, penalty mapping/cardinality, legal basis and runtime result.	Trigger violation from configured response and prove linked penalty/result/audit	Manual typing of violations/penalties disconnected from configured rules
ENG-09	Review, Return & Version Engine	Review, iPad, Web	Read-only submitted versions; approve/return/reject; immutable reviewer comments; selective unlock; new resubmission version; comparison.	Return two sections, prove all others remain locked, resubmit and compare versions	Editing original submitted version in place

ID	Engine	Consumers	Mandatory behavior	Acceptance proof	Forbidden shortcut
ENG-1 0	Offline Sync & Conflict Engine	iPad	Encrypted package/cache, autosave, crash recovery, queued operations, sync status, conflict handling, retry/idempotency and no silent overwrite.	Execute inspection offline, restart, resume, reconnect, sync and prove no data loss/duplication	Offline badge without real local persistence and recovery
ENG-1 1	Notification, SLA & Escalation Engine	All channels	Events, recipients, templates, reminders, SLA timers, overdue rules and escalations required by MVP1.	Demonstrate assignment, return, cancellation, expiry, appointment and overdue notifications	Hard-coded toasts with no event/recipient governance
ENG-1 2	Audit, Version & Non-Regression Traceability	All channels	Immutable event history, source requirement IDs, configuration versions, actor/time, before/after state, evidence links and acceptance proof.	Trace a source row from spreadsheet to implementation, runtime event, test and evidence	Marking complete without reproducible evidence

7. Cross-cutting controls and non-functional requirements

- Role-based access and least privilege apply to every screen, action and data scope.
- Every state-changing action must be auditable with actor, time, object, before/after state and version references where relevant.
- No submitted inspection, published configuration, reviewer decision or immutable audit record may be edited silently in place.
- Validation errors must preserve user work and identify exact blockers; partial state transitions are prohibited.
- All retryable operations must be idempotent so retries cannot create duplicate visits, evidence, notifications or submissions.
- The UI must use high-contrast, readable typography, strong hierarchy and consistent canonical components across all modules.
- Accessibility, localization readiness and responsive/device-specific behavior must be designed as system qualities rather than late-stage patches.
- Accepted behavior becomes part of the protected regression baseline.

8. iPad UI/UX authority

- Do not shrink the web portal. Build a field-first touch experience with large targets, strong daylight contrast, minimal navigation depth and persistent offline/autosave/readiness status.
- Primary execution must use clear progressive sections and unmistakable mandatory versus optional states.
- Camera/evidence controls, check-in, exception and submission actions must be optimized for touch and interruption recovery.
- Show blockers before the final submit moment; do not surprise the inspector with hidden missing requirements after a long field visit.
- Design both portrait and landscape behavior deliberately for the supported iPad devices; avoid accidental responsive collapse.

9. Complete channel requirement catalogue

This appendix preserves all 238 spreadsheet requirement rows mapped to this functional specification. The acceptance workbook remains the machine-readable execution contract.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M03-00 1	Inspector Dashboard	View workload, KPIs, quick actions, notifications.	Load dashboard widgets and inspector summary.	Dashboard displays inspector-specific information only.	Inspector can access dashboard and view assigned workload.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M03-002	View Assigned Visits	View visits, open visit, switch between views.	Load assigned visits and synchronize data.	Only assigned visits are displayed.	Inspector can access all assigned visits.
MVP1-M03-003	Calendar, List & Map Views	Switch between Calendar, List and Map.	Display selected view.	New visits appear only in Assignment Pool. Prepared visits appear in Calendar. Map displays all assigned visits.	Inspector can switch between all views.
MVP1-M03-004	Search, Filter & Sort	Search, filter, sort.	Apply search and filtering.	Search applies across all assigned visits.	Accurate search and filtering results.
MVP1-M03-005	Manage My Schedule	Drag & drop prepared visits between eligible dates.	Save updated schedule and refresh calendar.	Execution Date must remain within Planning Window.	Calendar updates successfully after changes.
MVP1-M03-006	Return Assignment	Select return reason and comments.	Update assignment and notify planner.	Return reason is mandatory.	Assignment successfully returned.
MVP1-M03-007	View Visit Card	Review visit details.	Load visit information.	Visit information is read-only.	Visit information displayed correctly.
MVP1-M03-008	View Factory 360°	Review factory profile and history.	Load factory information, history and location.	Latest synchronized factory data shall be displayed.	Factory profile loads successfully.
MVP1-M03-009	AI Preparation Assistant	Review AI insights.	Generate AI recommendations.	AI recommendations are advisory only.	AI recommendations generated successfully.
MVP1-M03-010	Prepare Inspection	Select Execution Date, Report Type, Execution Mode, Notify Factory, Save.	Validate inputs, resolve package, update status, populate calendar.	Preparation required before inspection can start.	Inspection successfully prepared.
MVP1-M03-011	Execution Mode Eligibility	Select eligible execution mode.	Evaluate eligibility rules.	Only eligible execution modes are displayed.	Ineligible modes cannot be selected.
MVP1-M03-012	Inspection Package Resolution	Review package preview.	Load Form, Checklist, Action Forms, Validation Rules, Required Evidence, Report Template and lock package.	Package regenerated after edits before execution and locked after save.	Correct inspection package generated.
MVP1-M03-013	Notify Factory	Enable/Disable notification.	Send notification using configured channels.	Optional for Physical inspections. Virtual notifications follow appointment workflow.	Notification successfully sent when enabled.
MVP1-M03-014	Virtual Appointment Management	Schedule appointment, review confirmations, approve reschedule requests.	Notify factory, receive confirmation, generate meeting, synchronize calendars.	Virtual inspection requires confirmed appointment before execution.	Virtual appointment confirmed successfully.
MVP1-M03-015	Inspection Status Management	-	Manage transitions between New, Ready and Expired.	Status lifecycle follows business rules and automatic expiration policies.	Status transitions occur correctly.
MVP1-M04-001	Validate Assigned Visit	Click Start Journey.	Verify the visit is assigned to the logged-in inspector.	Only the assigned inspector can start the journey. Shared or reassigned visits shall follow the latest assignment.	Inspector cannot start a visit assigned to another inspector.
MVP1-M04-002	Validate Visit Status	Click Start Journey.	Validate visit is Published and not Cancelled, Completed or Expired.	Journey can only start for executable visits.	Invalid visit statuses are blocked with a clear error message.
MVP1-M04-003	Validate Execution Window	Click Start Journey.	Compare current date/time against planned execution window.	Journey cannot start after the visit has expired.	Inspector receives appropriate validation if outside execution window.
MVP1-M04-004	Validate Factory Location	Click Start Journey.	Verify that the factory has valid coordinates. If planner manually selected location, use planner location; otherwise use license location.	Journey cannot start without a destination.	Navigation starts using the correct destination.
MVP1-M04-005	Download Locked Inspection Package	Wait.	Download the latest approved package including factory information, inspection report, inspection items, previous answers, and configuration. Lock the downloaded version.	Package version is frozen after download. Future changes by planners shall not affect the ongoing inspection.	Package downloads successfully before journey begins.
MVP1-M04-006	Verify Package Integrity	None	Validate package version, checksum, mandatory files and configuration.	Inspection cannot continue using a corrupted package.	Invalid packages are rejected and re-downloaded.
MVP1-M04-007	Cache Offline Package	None	Store package securely on the device.	Package remains available while offline until inspection is submitted.	Inspector can continue execution without network connectivity.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-008	Create Journey Session	None	Create journey session linked to visit.	One active journey per visit.	Journey session created successfully.
MVP1-M04-009	Capture Journey Start Time	None	Capture server timestamp and device timestamp.	Server time is the official audit time.	Journey start time recorded.
MVP1-M04-010	Capture Journey Start GPS	None	Capture latitude, longitude and altitude.	Coordinates stored as audit evidence.	GPS stored successfully.
MVP1-M04-011	Capture GPS Accuracy	None	Capture GPS accuracy in meters.	Used later for audit and investigation.	GPS accuracy stored.
MVP1-M04-012	Capture Device Information	None	Capture device ID, OS version, application version.	Every inspection must be traceable to a device.	Device information stored successfully.
MVP1-M04-013	Capture Network Status	None	Capture Online/Offline status and connection type.	Used for troubleshooting offline scenarios.	Initial connectivity stored.
MVP1-M04-014	Start Live Tracking	None	Start background GPS tracking according to configured interval.	Tracking continues until inspection begins or journey is cancelled.	Operations Center receives live location updates.
MVP1-M04-015	Initialize Journey Timeline	None	Create timeline events starting with Journey Started.	Every operational event must appear in timeline.	Timeline initialized successfully.
MVP1-M04-016	Launch Navigation	Open Navigation.	Launch preferred navigation application using factory coordinates.	Navigation is optional but recommended.	Inspector navigates to correct destination.
MVP1-M04-017	Calculate Initial ETA	None	Calculate ETA based on current location and destination.	ETA updates continuously during journey.	ETA displayed correctly.
MVP1-M04-018	Update Operational State	None	Change Operational State from Prepared to On the Way.	Workflow Status remains unchanged until inspection begins.	Operational State updated correctly.
MVP1-M04-019	Generate Audit Events	None	Log every significant event including validation, download, GPS capture, timeline creation and tracking initialization.	Every event must be auditable.	Complete audit trail available.
MVP1-M04-020	Error Handling	Retry / Cancel.	Display clear error messages, retry transient failures, prevent journey start if critical validations fail.	No partial journey session shall remain active after critical failure.	User receives meaningful error messages and system remains in a consistent state.
MVP1-M04-021	Activate Live Journey Tracking	None	Start continuous GPS tracking based on configurable interval.	Tracking starts only after successful Start Journey.	Live tracking starts successfully.
MVP1-M04-022	Capture Live GPS Coordinates	None	Capture Latitude, Longitude, Altitude, Speed, Heading and GPS Accuracy at each tracking interval.	Every GPS point is linked to the journey session.	GPS history is stored successfully.
MVP1-M04-023	Update Inspector Location on Operations Map	None	Update inspector marker on Operations Center map.	Map refresh interval shall be configurable.	Operations users can monitor inspector movement.
MVP1-M04-024	Calculate Estimated Time of Arrival (ETA)	None	Calculate ETA based on current location, road network and destination. Refresh ETA periodically.	ETA shall refresh automatically during travel.	ETA displayed accurately.
MVP1-M04-025	Calculate Remaining Distance	None	Calculate remaining travel distance using navigation route.	Distance refreshes automatically.	Remaining distance displayed correctly.
MVP1-M04-026	Monitor Journey Progress	None	Calculate journey progress percentage using travelled vs remaining distance.	Progress is informational only.	Progress updates correctly.
MVP1-M04-027	Monitor GPS Accuracy	None	Continuously monitor GPS accuracy and record significant degradation.	GPS quality shall be auditable.	GPS quality recorded correctly.
MVP1-M04-028	Monitor Network Connectivity	None	Detect Online, Offline, Wi-Fi and Mobile Network transitions.	Every connectivity change shall be recorded.	Connectivity status updates correctly.
MVP1-M04-029	Handle Offline Mode	Continue travelling.	Continue tracking locally, queue journey events, cache GPS updates until connectivity returns.	Journey shall continue without internet connectivity.	Inspector continues working without interruption.
MVP1-M04-030	Synchronize Offline Journey Data	None	Synchronize queued GPS points, timeline events and audit logs in chronological order.	Duplicate synchronization shall be prevented.	Queued data synchronized successfully.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-031	Monitor Route Deviation	None	Compare current location against planned navigation route.	Route deviation does not stop the inspection.	Deviation events recorded.
MVP1-M04-032	Detect Arrival Radius	None	Compare current location against configured arrival radius.	Arrival button becomes available only when within configured radius or GPS override is allowed.	Arrival eligibility evaluated correctly.
MVP1-M04-033	Record Journey Timeline Events	None	Record Journey Started, Tracking Started, Offline, Online, ETA Updates, Route Deviations and Arrival Detection.	Timeline must remain immutable after submission.	Timeline contains all operational events.
MVP1-M04-034	Generate Operational Audit Logs	None	Record GPS events, network changes, synchronization events and operational state changes.	All system-generated actions shall be auditable.	Complete audit trail available.
MVP1-M04-035	Display Journey Status	View status.	Display current Operational State, ETA, Distance Remaining, GPS Status, Connectivity Status and Progress.	Information refreshes automatically.	Journey dashboard displays current operational status.
MVP1-M04-036	Handle Journey Exceptions	Retry if prompted.	Handle GPS unavailable, navigation failure, synchronization failure, connectivity failure and mapping service errors. Maintain journey consistency and prevent duplicate sessions.	Critical failures shall never corrupt the active journey session.	Journey remains recoverable after unexpected failures.
MVP1-M04-037	Detect Arrival Radius	Continue travelling.	Continuously compare inspector GPS against the planned factory location and determine whether the inspector is inside the configured arrival radius.	Arrival radius shall be configurable by system administration.	System correctly identifies when the inspector enters the allowed arrival radius.
MVP1-M04-038	Enable Arrival Action	Click Arrived.	Enable the Arrived action only after arrival conditions are met or when GPS override is permitted.	Arrived button shall not be enabled before reaching the configured radius unless override is allowed.	Inspector can only continue when arrival conditions are satisfied.
MVP1-M04-039	Capture Arrival Timestamp	Confirm Arrival.	Capture both server timestamp and device timestamp as official arrival evidence.	Server timestamp shall be the official audit timestamp.	Arrival time is recorded successfully.
MVP1-M04-040	Capture Actual GPS Coordinates	Confirm Arrival.	Record latitude, longitude, altitude and GPS accuracy at the moment of arrival.	Actual coordinates shall always be stored regardless of GPS validation outcome.	Actual coordinates are successfully stored.
MVP1-M04-041	Compare Planned vs Actual Location	None	Calculate the distance between the planned location and the inspector's actual location.	Planned location is the Planner-selected location if available; otherwise, the Industrial License location from Senaei.	System calculates the distance correctly.
MVP1-M04-042	Validate GPS Distance	None	Compare calculated distance against the configured tolerance radius.	Validation shall use configurable radius values.	Validation result is generated correctly.
MVP1-M04-043	GPS Override Confirmation	Confirm Override or Cancel.	Display a confirmation dialog explaining that the inspector is outside the planned location. If confirmed, continue using the actual coordinates.	GPS override requires explicit inspector confirmation.	Override confirmation is recorded before continuing.
MVP1-M04-044	Store Actual Inspection Location	None	Save the final inspection coordinates that will be associated with the inspection execution.	Final coordinates shall remain immutable after inspection starts.	Inspection location stored successfully.
MVP1-M04-045	Capture Arrival Evidence	Capture photo or comment (if required).	Store arrival evidence and associate it with the arrival event.	Arrival evidence requirements shall be configurable.	Required evidence is successfully attached.
MVP1-M04-046	Update Operational State	None	Update Operational State from On the Way to Arrived while keeping Workflow Status unchanged.	Workflow Status remains unchanged until the inspection officially starts.	Operational State updated correctly.
MVP1-M04-047	Generate Arrival Timeline Events	None	Record Arrived, GPS Validated, GPS Override (if applicable), Arrival Evidence and Arrival Confirmed events.	Timeline events shall be immutable after submission.	Timeline accurately reflects the arrival process.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-048	Generate Audit Logs	None	Log timestamps, coordinates, validation results, overrides, evidence uploads and operational state transitions.	Every arrival-related action shall be auditable.	Complete audit trail is available.
MVP1-M04-049	Handle Arrival Validation Errors	Retry, Refresh GPS or Cancel.	Handle GPS unavailable, inaccurate GPS, mapping service failures and validation exceptions without corrupting the active inspection session.	Critical failures shall never automatically cancel the visit. Inspector shall always receive a clear validation message and available recovery options.	Arrival validation errors are handled without data loss or workflow inconsistency.
MVP1-M04-050	Load Arrival Confirmation Screen	None	Load Arrival Confirmation screen. Display Factory Card, Visit Card, Arrival Summary and available actions.	Arrival Confirmation is only accessible after successful arrival validation.	Screen loads successfully with all required information.
MVP1-M04-051	Load Factory Card	View Factory Card.	Load Factory Card summary including Factory Name, CR Number, License Number, License Status, Factory Status, Activity, Risk Level and Factory Health Score (if available).	Factory Card is read-only during Arrival Confirmation.	Factory Card displays the latest synchronized information.
MVP1-M04-052	Load Visit Card	View Visit Card.	Load Visit Type, Planned Window, Inspector Name, Branch, Planner, Visit Reason, Visit Priority and Report Type.	Visit Card reflects the locked visit package.	Visit Card displays correctly.
MVP1-M04-053	Display Journey Summary	Review information.	Display Journey Start Time, Arrival Time, Journey Duration, Distance Travelled and GPS Validation Result.	Journey Summary is informational only.	Journey summary displays correctly.
MVP1-M04-054	Display Context Panel	Expand or collapse Factory Card and Visit Card.	Provide quick access to Factory Card, Visit Card and map without leaving the confirmation screen.	Context panel remains available throughout execution.	Context panel functions correctly.
MVP1-M04-055	Proceed with Inspection	Click Proceed with Inspection.	Change Workflow Status to In Progress. Change Operational State from Arrived to Executing. Create Inspection Session. Initialize Execution Workspace. Stop Journey Tracking. Start Inspection Timeline. Start Inspection SLA.	Inspection can only start after successful arrival confirmation.	Execution workspace opens successfully.
MVP1-M04-056	Cancel Visit	Click Cancel Visit.	Display cancellation form and stop transition to execution.	Cancellation is only allowed before inspection starts.	Cancellation process starts correctly.
MVP1-M04-057	Capture Cancellation Reason	Select reason and enter comments if required.	Load configurable cancellation reasons. Save selected reason and comments.	Cancellation reason is mandatory. Comments may be mandatory depending on selected reason.	Cancellation cannot continue without mandatory information.
MVP1-M04-058	Capture Cancellation Evidence	Upload photo, video, document or comments.	Store evidence and link it to the cancellation record.	Evidence requirements are configurable per cancellation reason.	Mandatory evidence is enforced correctly.
MVP1-M04-059	Update Visit Status After Cancellation	None	Update Workflow Status to Cancelled. Update Operational State to Cancelled. Close active journey. Stop SLA timers.	Cancelled visits cannot be resumed.	Visit status updates correctly.
MVP1-M04-060	Generate Arrival Audit Events	None	Log Factory Card viewed, Visit Card viewed, Proceed with Inspection, Cancel Visit, Cancellation Reason, Cancellation Evidence and Workflow Transition.	Every action shall be auditable.	Complete audit trail is generated.
MVP1-M04-061	Handle Arrival Confirmation Errors	Retry or Return.	Handle workflow failures, inspection session creation failures, package initialization failures and cancellation failures without leaving the visit in an inconsistent state.	The visit shall never be partially transitioned between Journey and Execution. Recovery must always preserve data integrity.	Errors are handled gracefully and no inconsistent workflow state exists.
MVP1-M04-062	Create Inspection Session	Click Proceed with Inspection.	Create a unique Inspection Session linked to the Visit, Factory, Inspector and Package Version.	Only one active inspection session may exist per visit.	Inspection Session created successfully.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-063	Lock Inspection Package	None	Lock the downloaded package including Factory Information, Report, Inspection Items, Configuration and Rules.	Planner changes after execution starts shall not affect the active inspection.	Package remains unchanged throughout execution.
MVP1-M04-064	Load Execution Workspace	None	Load the Execution Workspace and initialize all required components.	Workspace shall only open after successful initialization.	Workspace loads successfully.
MVP1-M04-065	Initialize Wizard	None	Create wizard instance, calculate available steps and initialize navigation.	Wizard sequence shall follow the configured inspection flow.	Wizard initialized successfully.
MVP1-M04-066	Load Wizard Steps	None	Load Factory Information, Inspection Report, Inspection Items, Action Forms and Review steps. Hide Action Forms if no forms are generated later.	Wizard steps are configuration-driven.	Correct wizard sequence displayed.
MVP1-M04-067	Load Progress Engine	None	Calculate initial progress and display completion indicator.	Progress starts at 0% and updates automatically.	Progress displayed correctly.
MVP1-M04-068	Load Context Panel	Open/Collapse Context Panel.	Load Factory Card and Visit Card as a collapsible panel accessible from every wizard step.	Context panel remains available throughout execution.	Inspector can access context from any step.
MVP1-M04-069	Restore Draft Session	Resume Draft.	Restore the last completed wizard step, entered values, uploaded evidence and pending action forms.	Only the assigned inspector can resume the draft.	Draft restored successfully.
MVP1-M04-070	Initialize Auto Save	None	Start automatic background save based on configurable interval and significant user actions.	Auto Save shall not interrupt the inspector.	Inspection progress saved automatically.
MVP1-M04-071	Initialize Offline Mode	None	Verify offline package availability and initialize local storage for execution data.	Inspector shall continue working without network connectivity.	Offline execution initialized successfully.
MVP1-M04-072	Start Inspection Timeline	None	Create inspection timeline starting with Execution Started.	Timeline continues until inspection submission.	Timeline initialized correctly.
MVP1-M04-073	Start Inspection SLA	None	Capture execution start time and start SLA timer.	SLA starts when execution begins, not when journey starts.	SLA timer started successfully.
MVP1-M04-074	Validate Workspace Initialization	None	Validate package integrity, wizard configuration, mandatory forms and dependencies before allowing execution.	Inspection cannot continue if initialization fails.	Workspace validation completed successfully.
MVP1-M04-075	Handle Workspace Initialization Errors	Retry Initialization.	Handle package loading failures, missing configurations, wizard initialization failures and dependency errors. Maintain session consistency.	Failed initialization shall never create a partial inspection session.	Execution either initializes completely or fails safely.
MVP1-M04-076	Load Wizard Structure	None	Load all configured wizard steps from the Dynamic Form Engine in the configured sequence.	Wizard sequence shall be configuration-driven.	Wizard displays the correct steps.
MVP1-M04-077	Display Current Step	View current step.	Highlight current step and display step title and description.	Only one active step can exist at a time.	Active step is always visible.
MVP1-M04-078	Navigate to Next Step	Click Next.	Validate current step before moving to the next step.	Navigation only succeeds when validation passes.	User moves successfully to the next step.
MVP1-M04-079	Navigate to Previous Step	Click Previous.	Load previously entered data without data loss.	Previously entered data shall remain unchanged.	Previous step loads correctly.
MVP1-M04-080	Calculate Progress	None	Calculate completion based on Factory Information, Report, Inspection Items, Action Forms and Review.	Progress updates immediately after completing each section.	Progress reflects actual completion.
MVP1-M04-081	Display Progress Indicator	View progress.	Display percentage, completed steps and remaining steps.	Progress is informational only.	Progress indicator updates continuously.
MVP1-M04-082	Track Step Completion	Complete required fields.	Evaluate completion rules and mark the step as completed.	A step is complete only when mandatory requirements are satisfied.	Completed steps are clearly identified.
MVP1-M04-083	Validate Step Before Navigation	Click Next.	Validate mandatory fields, mandatory evidence and mandatory forms.	Navigation shall be blocked when validation fails.	User receives validation message.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-084	Skip Empty Action Forms Step	None	Automatically remove Action Forms step if no forms are generated.	Action Forms step appears only when required.	Wizard hides unnecessary steps.
MVP1-M04-085	Display Change Counter	View change summary.	Count Factory Information updates, Inspection Item updates and generated Action Forms.	Counter updates immediately after every modification.	Change counter always reflects actual changes.
MVP1-M04-086	Preserve Wizard State	None	Preserve active step, entered data and navigation history.	State shall survive refresh, reconnect and resume.	Wizard restores correctly after interruption.
MVP1-M04-087	Trigger Auto Save	None	Save progress automatically after configurable intervals and significant actions (field updates, evidence uploads, navigation, action form completion).	Auto Save shall run silently in the background.	No entered data is lost.
MVP1-M04-088	Save Draft	Click Save Draft.	Save current wizard state, values, evidence, generated forms and navigation position.	Draft remains editable by the assigned inspector only.	Draft restored exactly as saved.
MVP1-M04-089	Restore Draft Navigation	Resume Draft.	Open the wizard on the last active step with all saved data restored.	Resume shall not restart the wizard from Step 1.	Draft resumes successfully.
MVP1-M04-090	Display Validation Summary	View validation summary.	Display grouped validation messages by wizard step.	Validation summary shall identify the affected step.	Inspector can quickly identify missing information.
MVP1-M04-091	Jump to Missing Section	Click validation message.	Navigate directly to the corresponding wizard step and highlight the missing field.	Jump-to-Missing is available only for unresolved validations.	Inspector reaches the correct field immediately.
MVP1-M04-092	Lock Completed Navigation After Submission	View inspection.	Disable wizard navigation and editing after successful submission.	Submitted inspections become permanently read-only.	Inspector cannot edit after submission.
MVP1-M04-093	Generate Wizard Audit Events	None	Log step changes, draft saves, resumes, validation failures, navigation events and completion events.	Every wizard interaction shall be auditable.	Complete wizard audit history is available.
MVP1-M04-094	Handle Wizard Errors	Retry navigation.	Handle navigation failures, validation failures, draft failures and restore failures without data loss.	Navigation failures shall never corrupt the inspection session.	Wizard remains recoverable after unexpected failures.
MVP1-M04-095	Load Factory Information	Open Factory Information.	Load latest approved factory profile from Senaei.	Only synchronized Senaei data shall be displayed.	Factory information loads successfully.
MVP1-M04-096	Load Industrial License Information	Review information.	Load License Number, Status, Stage, Issue Date, Expiry Date, License Activity and License Holder.	License information is read-only.	License information displayed correctly.
MVP1-M04-097	Load Factory General Information	Review information.	Display Factory Name, CR Number, Factory Status, Activity, Region, City and Address.	Data originates from Senaei only.	Factory profile displayed correctly.
MVP1-M04-098	Load Products	Review products.	Load all registered products with HS Codes and production details.	Products are loaded from the license package.	Products displayed successfully.
MVP1-M04-099	Load Raw Materials	Review raw materials.	Load all synchronized raw materials.	Read-only until observation is recorded.	Raw materials displayed correctly.
MVP1-M04-100	Load Employees	Review employee numbers.	Load employee statistics from Senaei.	Source values remain unchanged during inspection.	Employee information displayed correctly.
MVP1-M04-101	Load Investment Information	Review investment.	Display investment size, production capacity and configured industrial indicators.	Read-only source information.	Investment information displayed correctly.
MVP1-M04-102	Display Source Value	Review value.	Display every attribute as Source Value (Senaei).	Source values cannot be edited directly.	Source values clearly identified.
MVP1-M04-103	Capture Observed Value	Enter observed value.	Save inspector observation separately from Senaei data.	Observation never overwrites source value.	Observation saved successfully.
MVP1-M04-104	Mark Field as Verified	Click Verified.	Mark field verification status as Verified.	Verified fields require no observed value.	Verified status recorded.
MVP1-M04-105	Mark Field as Updated	Enter new value.	Mark verification status as Updated.	Updated fields require observed value.	Updated status recorded correctly.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-106	Compare Before / After Values	Review values.	Display Source Value vs Observed Value side by side.	Every updated field shall display both values.	Before/After comparison available.
MVP1-M04-107	Highlight Updated Fields	Review highlights.	Highlight every updated field.	Highlights remain visible until submission.	Modified fields easily identifiable.
MVP1-M04-108	Attach Evidence to Factory Data	Upload Photo, Video, Document or Comment.	Associate evidence with the specific updated field.	Evidence can be attached to any verified field.	Evidence linked successfully.
MVP1-M04-109	Annotate Evidence	Draw annotations.	Save annotated image together with the original file.	Original evidence shall never be overwritten.	Annotated evidence stored successfully.
MVP1-M04-110	Maintain Change Counter	Review change counter.	Count all updated factory fields.	Counter updates immediately.	Counter always accurate.
MVP1-M04-111	Review Factory Changes	Review changes.	Display every updated field with Source Value, Observed Value and attached evidence.	Review is read-only.	Review summary complete.
MVP1-M04-112	Preserve Source Integrity	None	Prevent any direct update to Senaei during inspection.	Source data remains unchanged until future synchronization workflow.	Master data integrity maintained.
MVP1-M04-113	Generate Verification Audit	None	Log Verified, Updated, Evidence Added, Comments Added and Before/After values.	Every verification action shall be auditable.	Complete audit trail generated.
MVP1-M04-114	Handle Verification Errors	Retry.	Handle synchronization failures, invalid observations and evidence failures without losing entered data.	Verification failures shall never remove inspector observations.	Errors handled without data loss.
MVP1-M04-115	Load Inspection Report	Open Report Step.	Retrieve the report definition selected during Visit Preparation from the Dynamic Form Builder and render the latest locked version included in the inspection package.	Report definition is locked when the inspection starts and cannot change during execution.	Correct report loads successfully.
MVP1-M04-116	Load Report Metadata	Review report details.	Display Report Name, Version, Description, Inspection Type and Last Updated Date.	Report metadata is read-only.	Metadata displayed correctly.
MVP1-M04-117	Render Report Sections	Navigate between sections.	Render report sections exactly as configured in the Form Builder.	Section order follows configuration.	All configured sections displayed correctly.
MVP1-M04-118	Render Dynamic Fields	Fill field values.	Render all configured field types including text, number, date, lookup, checkbox, radio button, attachment, image, signature and custom controls.	Field behavior is driven entirely by configuration.	Dynamic fields render successfully.
MVP1-M04-119	Apply Field Visibility Rules	Complete report.	Evaluate conditional visibility rules configured in the Form Builder.	Hidden fields are excluded from validation.	Visibility rules applied correctly.
MVP1-M04-120	Apply Mandatory Rules	Complete mandatory fields.	Validate mandatory fields according to Form Builder configuration.	Mandatory rules are configuration-driven.	Missing mandatory fields block progression.
MVP1-M04-121	Apply Conditional Validation	Enter values.	Execute conditional validations, value ranges, dependencies and custom validation rules configured in the Form Builder.	Validation follows configured business rules only.	Conditional validations execute successfully.
MVP1-M04-122	Capture Report Responses	Enter values.	Store responses continuously as part of the active inspection session.	Responses belong to the current inspection version only.	Responses saved correctly.
MVP1-M04-123	Attach Evidence to Report Fields	Upload Photo, Video, Document or Comment.	Link uploaded evidence to the corresponding report field.	Evidence remains attached to the originating field.	Evidence linked successfully.
MVP1-M04-124	Annotate Report Evidence	Annotate image.	Store annotated version while preserving the original file.	Original image shall never be modified.	Annotation stored successfully.
MVP1-M04-125	Save Draft Report	Save Draft.	Save current responses, uploaded evidence and completed sections.	Draft remains editable until submission.	Draft restored successfully.
MVP1-M04-126	Auto Save Report	Continue working.	Automatically save report after configurable intervals and significant changes.	Auto Save runs silently in the background.	No entered data is lost.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-127	Calculate Report Completion	Complete report.	Calculate completed vs pending report fields and update overall wizard progress.	Completion contributes to overall inspection progress.	Completion calculated accurately.
MVP1-M04-128	Review Report Summary	Review summary.	Display completed sections, pending fields and attached evidence summary.	Summary is informational before submission.	Summary generated successfully.
MVP1-M04-129	Lock Report After Submission	View Report.	Render report in read-only mode after submission.	Report becomes immutable immediately after submission.	Report cannot be modified after submission.
MVP1-M04-130	Generate Report Audit Events	None	Record report opened, section completed, field updated, evidence uploaded, draft saved, resumed and completed events.	Every report interaction shall be auditable.	Audit trail contains all report activities.
MVP1-M04-131	Handle Report Errors	Retry.	Handle rendering failures, validation failures, save failures and evidence upload failures while preserving entered information.	Report failures shall never cause data loss or inconsistent inspection state.	Report recovers gracefully after failures.
MVP1-M04-132	Load Inspection Checklist	Open Checklist Step.	Load all configured inspection items from Compliance Configuration according to the selected Report Type.	Checklist is loaded from the locked inspection package.	Correct checklist is displayed.
MVP1-M04-133	Load Inspection Categories	Expand / Collapse Categories.	Display checklist grouped by configured categories and subcategories.	Category sequence follows configuration.	Categories displayed correctly.
MVP1-M04-134	Load Required Inspection Items	Answer Item.	Mark configured mandatory items.	Mandatory items cannot be skipped.	Mandatory items clearly identified.
MVP1-M04-135	Load Optional Inspection Items	Answer or Skip.	Display optional items.	Optional items may remain unanswered.	Optional items identified correctly.
MVP1-M04-136	Load Previous Inspection Answers	Review previous answer.	Load latest approved answer from previous inspections for comparison.	Historical answers are read-only.	Previous answers displayed successfully.
MVP1-M04-137	Load Previous Evidence	Review evidence.	Load previous inspection photos and attachments.	Historical evidence is read-only.	Historical evidence displayed correctly.
MVP1-M04-138	Display Inspection Item Details	Review item.	Display Item Name, Description, Regulation Reference, Evidence Requirements, Guidance Notes and Response Type.	Item details are configuration-driven.	Inspector has complete inspection context.
MVP1-M04-139	Capture Inspection Response	Select response.	Save selected response for the inspection item.	Only configured response values are allowed.	Response saved successfully.
MVP1-M04-140	Apply Response Validation	Complete response.	Validate response according to configured response type and business rules.	Invalid responses cannot be saved.	Validation executes correctly.
MVP1-M04-141	Evaluate Compliance Result	None	Evaluate whether response is Compliant or Non-Compliant based on configured rules.	Evaluation is automatic.	Compliance calculated correctly.
MVP1-M04-142	Generate Violation	None	Retrieve linked violation from Compliance Configuration.	Inspector cannot modify generated violation.	Correct violation generated automatically.
MVP1-M04-143	Display Penalty	Review penalty.	Display configured penalty linked to the generated violation.	Penalty is informational during inspection.	Correct penalty displayed.
MVP1-M04-144	Display Corrective Action	Review corrective action.	Display configured corrective action associated with the generated violation.	Corrective action is read-only.	Correct corrective action displayed.
MVP1-M04-145	Capture Inspector Comments	Enter comments.	Save inspector comments linked to the inspection item.	Comments remain editable until submission.	Comments stored successfully.
MVP1-M04-146	Attach Evidence to Inspection Item	Upload Photo, Video, Document or Comment.	Link uploaded evidence to the inspection item.	Evidence remains attached to originating inspection item.	Evidence linked correctly.
MVP1-M04-147	Annotate Inspection Evidence	Annotate image.	Save annotation while preserving original media.	Original evidence remains unchanged.	Annotation stored successfully.
MVP1-M04-148	Track Checklist Progress	View progress.	Calculate completed items versus remaining items.	Progress contributes to overall inspection completion.	Progress displayed accurately.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-149	Review Checklist Summary	Review summary.	Display total answered items, pending items, compliant items, non-compliant items, violations and evidence count.	Summary updates dynamically.	Summary reflects actual inspection state.
MVP1-M04-150	Save Checklist Draft	Save Draft.	Save responses, comments and evidence.	Draft remains editable before submission.	Draft restored successfully.
MVP1-M04-151	Lock Checklist After Submission	View checklist.	Render checklist in read-only mode after submission.	Checklist becomes immutable after submission.	Checklist cannot be modified after submission.
MVP1-M04-152	Generate Checklist Audit Events	None	Record item opened, answer changed, violation generated, evidence uploaded, comments added, draft saved and completed events.	Every checklist action shall be auditable.	Audit history complete.
MVP1-M04-153	Handle Checklist Errors	Retry.	Handle validation failures, save failures, evidence upload failures and synchronization failures while preserving entered information.	No checklist data shall be lost because of recoverable failures.	Checklist remains recoverable after failures.
MVP1-M04-154	Upload Evidence	Upload Photo, Video or Document.	Upload and associate evidence with the active inspection session.	Evidence can be uploaded from any execution step.	Evidence uploaded successfully.
MVP1-M04-155	Select Evidence Source	Select Camera, Gallery or File.	Display supported evidence sources.	Supported sources are configurable.	Selected source opened successfully.
MVP1-M04-156	Capture Photo	Take Photo.	Launch device camera and capture image.	Original photo retained.	Photo stored successfully.
MVP1-M04-157	Capture Video	Record Video.	Launch video recorder and capture video.	Maximum recording duration configurable.	Video uploaded successfully.
MVP1-M04-158	Upload Document	Select Document.	Upload supported document types.	Allowed extensions configured centrally.	Document uploaded successfully.
MVP1-M04-159	Add Evidence Comment	Enter Comment.	Store comment with evidence record.	Comment length configurable.	Comment stored successfully.
MVP1-M04-160	Annotate Image	Draw Annotation.	Save annotation layer while preserving original image.	Original evidence shall never be modified.	Annotation saved successfully.
MVP1-M04-161	Link Evidence	None	Link evidence to Factory Information, Report Field, Inspection Item, Violation or Action Form.	One evidence item may be linked to multiple entities if configured.	Evidence linked correctly.
MVP1-M04-162	Preview Evidence	Open Evidence.	Display preview according to file type.	Preview does not modify evidence.	Evidence preview available.
MVP1-M04-163	Replace Evidence	Replace File.	Archive previous version and upload replacement.	Replacement allowed only before submission.	Previous version retained in audit.
MVP1-M04-164	Delete Evidence	Delete Evidence.	Remove evidence from inspection package while recording audit event.	Deletion allowed only before submission.	Evidence removed successfully.
MVP1-M04-165	Validate Evidence	None	Validate mandatory evidence, file type, file size and supported formats.	Submission blocked if mandatory evidence is missing.	Validation performed successfully.
MVP1-M04-166	Compress Media	None	Compress images and videos before synchronization without compromising required quality.	Original quality policy configurable.	Media optimized successfully.
MVP1-M04-167	Offline Evidence Storage	Continue Upload.	Store evidence securely on device until synchronization.	Evidence remains available after application restart.	Offline evidence retained successfully.
MVP1-M04-168	Synchronize Evidence	None	Synchronize queued evidence while preserving relationships.	Synchronization order shall be maintained.	Offline evidence synchronized successfully.
MVP1-M04-169	Generate Evidence Audit	None	Record upload, replace, delete, preview, annotation and synchronization events.	Every evidence action shall be auditable.	Complete evidence history available.
MVP1-M04-170	Handle Evidence Errors	Retry Upload.	Handle upload failures, synchronization failures, storage failures and unsupported formats without losing existing evidence.	Evidence integrity maintained after failures.	No uploaded evidence is lost because of recoverable failures.
MVP1-M04-171	Evaluate Action Form Triggers	None	Evaluate configured trigger rules after each inspection response.	Trigger evaluation occurs immediately after each answer.	Action Form decision generated correctly.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-172	Generate Action Form	None	Create Action Form from Dynamic Form Builder using configured template.	Generated forms belong only to the current inspection.	Correct Action Form generated.
MVP1-M04-173	Assign Action Form Type	None	Assign configured form type (Corrective Action, Notice, Acknowledgment, Shutdown, etc.).	Form type is configuration-driven.	Correct form type assigned.
MVP1-M04-174	Load Action Form	Open Form.	Load dynamic fields and sections according to configuration.	Forms are loaded from the locked package version.	Form displayed successfully.
MVP1-M04-175	Render Dynamic Form Fields	Complete fields.	Render text, number, lookup, date, attachment, signature and custom controls.	Field rendering follows Form Builder configuration.	Fields rendered correctly.
MVP1-M04-176	Apply Conditional Visibility	Fill data.	Evaluate conditional display rules defined in the Form Builder.	Hidden fields are excluded from validation.	Visibility rules executed correctly.
MVP1-M04-177	Validate Mandatory Fields	Complete required fields.	Validate mandatory fields before allowing completion.	Mandatory rules are configuration-driven.	Missing required fields block completion.
MVP1-M04-178	Capture Action Form Responses	Enter values.	Save responses as part of the inspection session.	Responses remain editable until submission.	Responses stored successfully.
MVP1-M04-179	Attach Evidence to Action Form	Upload evidence.	Link uploaded evidence directly to the Action Form.	Evidence requirements depend on form configuration.	Evidence linked correctly.
MVP1-M04-180	Capture Digital Signature	Sign electronically.	Capture inspector signature and, when configured, factory representative signature. If refused, record refusal reason.	Signature requirement is configurable.	Required signatures captured successfully.
MVP1-M04-181	Save Draft Action Form	Save Draft.	Save partially completed Action Form including responses and evidence.	Draft remains editable until submission.	Draft restored successfully.
MVP1-M04-182	Auto Save Action Form	Continue working.	Save Action Form automatically after configurable intervals and major changes.	Auto Save runs silently.	Form progress preserved automatically.
MVP1-M04-183	Complete Action Form	Click Complete.	Validate all mandatory requirements and mark Action Form as completed.	Form cannot be marked complete with validation errors.	Completed status assigned correctly.
MVP1-M04-184	Update Wizard Progress	None	Update inspection progress based on completed Action Forms.	Progress updates automatically.	Wizard progress recalculated correctly.
MVP1-M04-185	Review Generated Action Forms	Review forms.	Display generated forms, completion status, evidence count and signature status.	Summary is read-only before submission.	Summary generated successfully.
MVP1-M04-186	Lock Action Forms After Submission	View forms.	Render all Action Forms in read-only mode.	Forms become immutable after submission.	Action Forms cannot be edited.
MVP1-M04-187	Generate Action Form Audit Events	None	Record generation, editing, evidence upload, signatures, completion, draft save, resume and submission events.	Every Action Form event is auditable.	Audit history complete.
MVP1-M04-188	Handle Action Form Errors	Retry.	Handle generation failures, rendering failures, validation failures, signature failures and save failures while preserving entered data.	No completed Action Form data shall be lost due to recoverable failures.	Action Forms recover successfully after failures.
MVP1-M04-189	Load Review Workspace	Open Review.	Load review dashboard containing all execution sections.	Review is always the last wizard step.	Review screen loads successfully.
MVP1-M04-190	Review Factory Information	Review changes.	Display Source Value, Observed Value, Verification Status, Before/After values and attached evidence for every modified field.	Only modified fields are highlighted.	Factory verification summary displayed correctly.
MVP1-M04-191	Review Inspection Report	Expand Report Section.	Display report sections, completed fields, attachments and comments.	Report is read-only during review.	Report summary displayed successfully.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-192	Review Inspection Checklist	Expand Checklist Section.	Display answered items, pending items, compliant items, non-compliant items and inspector comments.	Checklist reflects the latest saved responses.	Checklist summary displayed correctly.
MVP1-M04-193	Review Violations	Expand Violations Section.	Display violation classification, regulation, severity, corrective action and current status.	Violations are read-only and generated automatically.	Violations displayed correctly.
MVP1-M04-194	Review Penalties	Review penalties.	Display configured penalties associated with each generated violation.	Penalties cannot be modified by the inspector.	Penalties displayed correctly.
MVP1-M04-195	Review Action Forms	Expand Action Forms.	Display completion status, signatures, evidence count and validation status for every generated form.	Incomplete mandatory forms are highlighted.	Action Form summary displayed successfully.
MVP1-M04-196	Review Evidence	Preview Evidence.	Display all evidence grouped by Factory Verification, Report, Checklist and Action Forms.	Deleted evidence shall not appear.	Evidence summary displayed correctly.
MVP1-M04-197	Review Signatures	Review Signatures.	Display Inspector Signature, Factory Representative Signature or Refusal Reason where applicable.	Signature requirements depend on Action Form configuration.	Signature status displayed correctly.
MVP1-M04-198	Calculate Change Counter	View summary.	Count updated Factory Information, Report responses, Checklist responses, Evidence uploads and Action Forms.	Counter updates automatically until submission.	Counter reflects all changes.
MVP1-M04-199	Validate Mandatory Information	Click Validate.	Validate mandatory fields, mandatory checklist items, mandatory Action Forms, signatures and evidence.	Submission blocked if any mandatory requirement is missing.	Validation completed successfully.
MVP1-M04-200	Display Validation Summary	Review issues.	Display grouped validation results by wizard step with issue description and severity.	All validation failures are displayed together.	Validation summary generated correctly.
MVP1-M04-201	Jump to Missing Section	Click Validation Issue.	Navigate to the corresponding wizard step and highlight the missing field or section.	Navigation always opens the correct location.	Inspector reaches missing field directly.
MVP1-M04-202	Expand / Collapse Review Sections	Expand or Collapse Sections.	Allow review sections to be expanded or collapsed independently.	Expansion state does not affect data.	Sections expand and collapse correctly.
MVP1-M04-203	Save Review Draft	Save Draft.	Save review position together with the inspection draft.	Draft remains editable until submission.	Review draft restored successfully.
MVP1-M04-204	Determine Submission Readiness	None	Evaluate overall inspection readiness based on validation results. Enable Submit only when all blocking validations pass.	Submit button remains disabled while blocking issues exist.	Submit button enabled only when inspection is ready.
MVP1-M04-205	Generate Review Audit Events	None	Record review opened, validation executed, navigation events, draft saved and submission readiness evaluation.	Every review activity shall be auditable.	Audit history complete.
MVP1-M04-206	Handle Review Errors	Retry Review.	Handle validation failures, summary generation failures and navigation failures while preserving inspection data.	Review errors shall never modify or remove inspection data.	Review remains recoverable without data loss.
MVP1-M04-207	Submit Inspection	Click Submit Inspection.	Start final submission process.	Submit is only available after Review validation passes.	Submission process starts successfully.
MVP1-M04-208	Execute Final Validation	None	Revalidate mandatory fields, mandatory checklist items, Action Forms, evidence, signatures and business rules before committing submission.	Final validation always executes even if previous validation passed.	Submission blocked if validation fails.
MVP1-M04-209	Generate Inspection Reference Number	None	Generate a unique Inspection Number according to configured numbering rules.	Inspection Number is immutable after generation.	Unique inspection number generated successfully.
MVP1-M04-210	Freeze Inspection Data	None	Freeze all responses, checklist answers, report data, factory observations, Action Forms and evidence.	No further editing is permitted after submission.	Inspection data permanently locked.

Requirement	Capability	User action	System reaction	Business rule	Acceptance
MVP1-M04-211	Lock Inspection Package	None	Lock the inspection package version used during execution for future audit and review.	Package version shall remain available throughout the inspection lifecycle.	Package version preserved successfully.
MVP1-M04-212	Finalize Evidence Repository	None	Lock uploaded evidence, annotations and comments against further modification.	Evidence becomes read-only after submission.	Evidence repository finalized successfully.
MVP1-M04-213	Finalize Action Forms	None	Mark all completed Action Forms as closed and prevent further editing.	All mandatory Action Forms must be completed before submission.	Action Forms finalized successfully.
MVP1-M04-214	Generate Inspection Snapshot	None	Create a read-only snapshot including package version, responses, observations, evidence, violations and Action Forms.	Snapshot becomes the official inspection record.	Snapshot created successfully.
MVP1-M04-215	Generate Inspection PDF	None	Generate PDF containing factory information, inspection report, checklist, violations, penalties, Action Forms, evidence references and inspector information.	PDF is generated after successful submission and becomes the official inspection report for reviewers.	PDF generated successfully.
MVP1-M04-216	Update Workflow Status	None	Update Workflow Status from In Progress to Pending Review.	Workflow transition is automatic.	Workflow updated successfully.
MVP1-M04-217	Update Operational State	None	Update Operational State from Executing to Submitted.	Operational State reflects execution completion.	Operational State updated successfully.
MVP1-M04-218	Assign Reviewer	None	Create reviewer work item according to workflow configuration and assignment rules.	Assignment follows workflow configuration.	Reviewer task created successfully.
MVP1-M04-219	Notify Reviewer	None	Send system notification to assigned reviewer.	Notification templates are configurable.	Reviewer receives notification successfully.
MVP1-M04-220	Notify Inspector	None	Display confirmation message and record submission completion.	Confirmation displayed immediately after successful submission.	Inspector receives confirmation successfully.
MVP1-M04-221	Convert Inspection to Read-Only	View Inspection.	Disable editing across all execution screens while allowing viewing of submitted information.	Inspector can no longer modify submitted inspection.	Inspection becomes completely read-only.
MVP1-M04-222	Generate Submission Audit Events	None	Record submission request, validation results, inspection number generation, package lock, PDF generation, workflow transition, reviewer assignment and notifications.	Every submission activity shall be permanently auditable.	Complete submission audit history available.
MVP1-M04-223	Handle Submission Failures	Retry Submission.	Handle workflow failures, PDF failures, notification failures, repository failures and assignment failures while ensuring the inspection is either completely submitted or completely rolled back.	Partial submission shall never occur. Data integrity shall always be maintained.	Submission failures recover safely without corrupting the inspection.